

Introduction to DBMS : → CRUD
Create Read Update Delete

* Database Management System

A DBMS is a software used to store, manage & retrieve data "efficiently".

- ✓ * MySQL → we are using this
 - ✓ * Oracle
 - * MongoDB
 - * PostgreSQL
- [Just like we write Java in Eclipse or IntelliJ]

Why DBMS : →

- * Reduces Redundancy (Duplicates)
- * Ensures data consistency
- * Provides security
- * Multi-user access (servers)

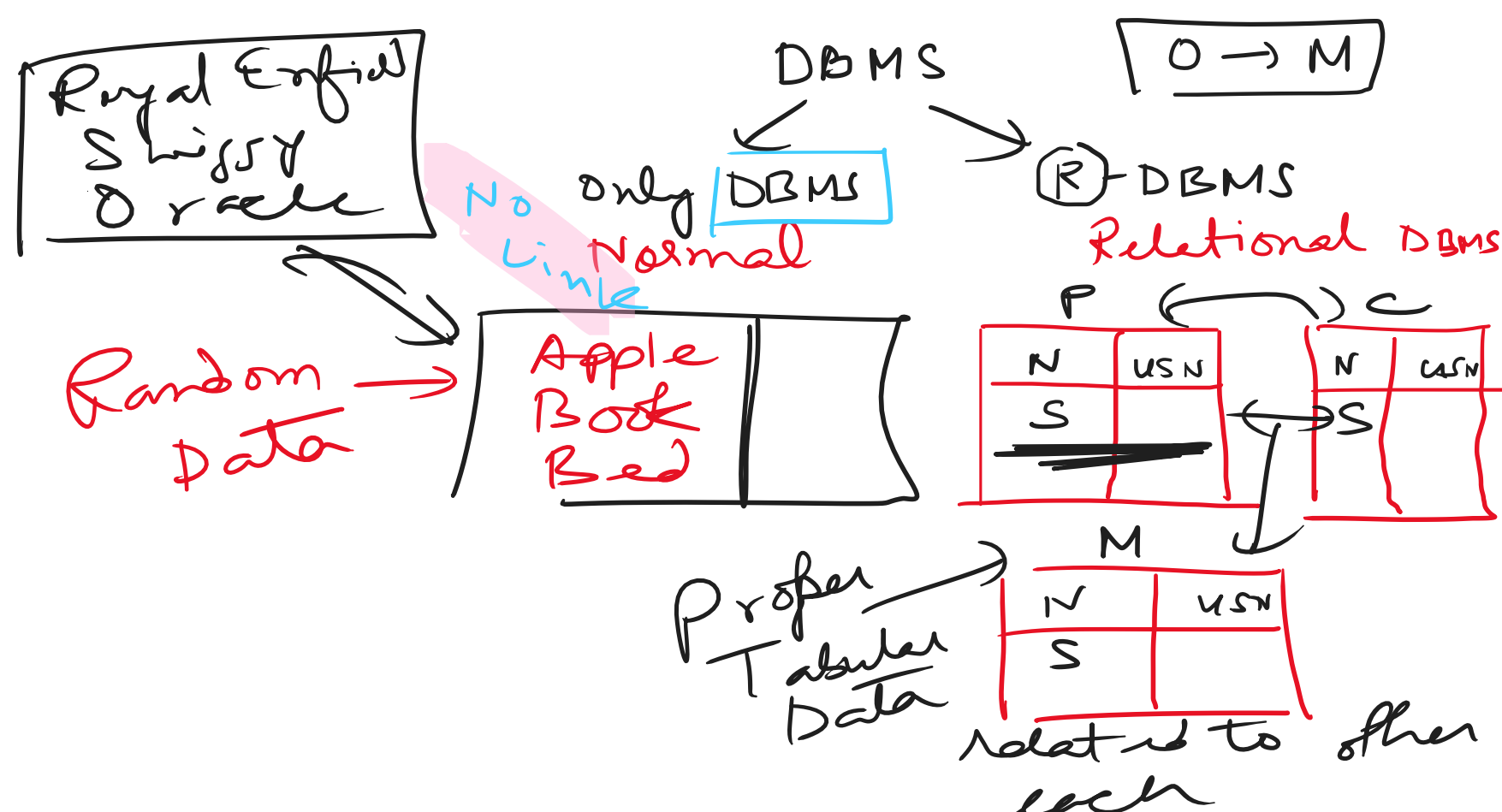
What is SQL : → (Sequel) (⇒)

Structure Query Language is used to interact with databases.

Uses : →

- (I) Creating Tables
- (II) Inserting Data
- (III) Retrieving Data
- (IV) Update/Delete Data

[CRUD] *****



- * If tables are related/connected → RDBMS
- * If not related/connected → DBMS

Types of SQL Languages

Type	Full Form	Purpose	Commands
DDL	Data Definition Language	Structure	Create, Alter, Drop, Truncate
DML	Data Manipulation Language	Data	Insert, Update, Delete
DQL	Data Query Language	Retrieve	Select
DCL	Data Control Language	Permissions	GRANT, REVOKE
TCL	Transaction Control Language	Transactions	COMMIT, ROLLBACK, SAVE POINT

Datatypes in SQL : →

Category	Data Type	Description	Example
Numeric	INT	Whole Numbers	10
Numeric	FLOAT	Decimal Numbers	10.5
String	VARCHAR(n)	Variable Text	'SNPSU'
String	CHAR(n)	Fixed Text	'A'
DATE	DATE	Date only	'2026-04-02'
Boolean	BOOLEAN	True/False	TRUE
<u>Special ones</u>			
Time & Date	TIMESTAMP	Both Date & Time	'2026-04-02 00:15:26'
Images	BLOB	Binary Large Object	(x.png)